# SRINIVAS R C

**AI/ML Engineer**

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## EDUCATION

**B.Tech in Artificial Intelligence and Machine Learning** | CGPA: 8.0/10 **2023 – 2027**

*REVA University, Bengaluru*

**PUC (PCMB)** | MES College of Arts, Science and Commerce, Bengaluru | 82.5% **2023**

**ICSE, Class X** | Mount Senoria School, Bengaluru | 88.8% **2021**

## EXPERIENCE

**AI Engineer Intern | VLED Lab, IIT Ropar | Remote** **Aug 2026 – Oct 2026**

*Project: Ajrasakha | React, TypeScript, Node.js, Express, MongoDB Atlas Vector Search, Sarvam AI, Ollama*

- Contributing to Ajrasakha, an open-source AI assistant that answers farmers' questions in their own language.
- Working on a 3-tier retrieval pipeline that serves expert-verified answers first, then the Package of Practices database, and calls local LLMs (DeepSeek-R1, Qwen3, GPT-OSS) through Ollama only as a fallback.
- Selected through a competitive evaluation; committing code as an assigned contributor on a live public repository.

## PROJECTS

**AEGIS: Air-Gapped Agentic AI Workbench | GitHub** **Sep 2026**

*Smart India Hackathon 2026 (PS SIH26117) | Python, LangGraph, Ollama, Qwen2.5, FAISS, BM25, Docker, Streamlit*

- Built an offline agentic AI assistant for Mangalore Refinery (MRPL) operators that answers questions from plant SOPs with no cloud calls or API keys, running a 7B model on a 4 GB laptop GPU.
- Structured the agent as a 7-node LangGraph state machine (Plan, Vision, RAG, Calc, Safety Check, Reflect, Answer) with hybrid FAISS and BM25 retrieval, merged through reciprocal-rank fusion to match exact SOP IDs.
- Kept safety verdicts out of the LLM: limits are checked in deterministic code, critical findings need a named supervisor sign-off, and every step is recorded in a SHA-256 hash-chained audit log.
- Sandboxed calculations in Docker and shipped one-click installers for Windows, Linux, and macOS.

**Codebase-Wide Version Migration and Refactoring Agent | GitHub** **Aug 2026 – Present**

*Major Project, Center for AI and ML (CAIML), REVA University | Python, LLM Agents, AST, Static Analysis, Docker*

- Building an autonomous LLM agent that upgrades entire codebases across breaking library-version changes.
- Mapping each codebase with AST-based static analysis and a dependency graph, then running a plan, edit, test, and self-correct loop inside a Docker sandbox.
- Producing a patch and audit log for each change, plus a benchmark report and a research paper.

**ArchAgent: AI Architectural Design Platform | GitHub** **Apr 2026 – Jun 2026**

*React, TypeScript, Google Gemini, Hugging Face FLUX, Three.js, Supabase, jsPDF*

- Built a platform that turns a plain-language building brief into 3D panoramic renders and an itemised INR cost estimate in under 60 seconds.
- Chained 4 Google Gemini prompt stages and paired them with Hugging Face FLUX image generation and an interactive Three.js 3D viewer.
- Reduced hallucinated cost estimates by grounding the LLM with few-shot INR price examples, and added Supabase authentication and 1-click PDF reports.

**AquaSentinel: Underwater Inspection Robot Platform | Live / GitHub** **Sep 2026**

*React 19, TypeScript, Vite, Node.js, Express, PostgreSQL (Neon), Computer Vision*

- Created a live mission-control dashboard with 5 core features: sensor telemetry, robot health monitoring, waypoint route planning, AI image defect detection, and PDF inspection reports.
- Wrote a 13-document engineering suite (PRD, API contract, database schema, 40+ tickets, security spec, CI workflow) and corrected a physics error in the legacy specs: GPS does not work underwater.

**Geo Measure: Geospatial File Measurement API** | **Live** / **GitHub** **Oct 2026**

*Python, FastAPI, React, PostgreSQL (Neon), Alembic, Docker, Render*

- Developed a REST API that accepts Shapefile (.zip) and KML uploads, extracts each feature's geometry, CRS, and properties, and returns the area of polygons and the length of lines.
- Reprojected every feature to its local UTM zone before measuring and cross-checked results with a geodesic calculation, flagging differences above 0.5%; naive Web Mercator overstates area by 5.8% near Bengaluru.
- Deployed the frontend and API as a single Docker image on Render, with measurement history stored in PostgreSQL and managed through Alembic migrations.

**AI Travel Planner** | **GitHub** **Jun 2026**

*React, Vite, Tailwind CSS, shadcn/ui, Google Gemini, Firebase Firestore, Google OAuth, Google Places API, React Leaflet*

- Developed an AI travel app that generates personalised day-by-day itineraries from 3 inputs (destination, budget, and preferences) using structured JSON prompts with Google Gemini.
- Integrated 4 Google services (Gemini, OAuth, Places API, and Firebase Firestore) for secure sign-in, live place data, saved trips, and React Leaflet maps of every stop.

**AI Finance Assistant (Team Project)** | **GitHub** **2026**

*Next.js, React, Prisma, Inngest, Arcjet, Tailwind CSS*

- Co-developed a Next.js personal finance assistant that explains financial concepts through prompt-engineered LLM responses, with dashboard, portfolio, and transaction modules.
- Designed a relational Prisma schema for accounts, transactions, and stock portfolios behind REST API routes, with Inngest serverless functions running background jobs.

## TECHNICAL SKILLS

**Languages:** Python, Java, C, C++, JavaScript, TypeScript, SQL, Bash

**AI and ML:** Machine Learning, Deep Learning, NLP, Computer Vision, Agentic AI, LLMs, RAG, LangGraph, Prompt Engineering, Hybrid Retrieval (FAISS, BM25), Vector Search, Ollama, Google Gemini API, Hugging Face, Scikit-learn, TensorFlow, Pandas, NumPy

**Backend and Databases:** FastAPI, Node.js, Express, REST APIs, PostgreSQL, MongoDB, SQLite, Prisma

**Cloud and Deployment:** Render, Vercel, Supabase, Firebase, Neon

**Frontend:** React, Next.js, Vite, Tailwind CSS, Three.js, Streamlit

**Operating Systems:** Windows, Linux (Fedora, Pop!_OS, Arch Linux)

**Tools:** Git, GitHub, GitHub Actions (CI/CD Workflows), Docker, VS Code, Jupyter Notebook

**Core CS:** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks

## CERTIFICATIONS

**Artificial Intelligence: Foundations and Algorithms** | IIT Delhi / NPTEL **2026**

**Deep Learning** (12-week, proctored) | IIT Ropar / NPTEL **2026**

**Software Engineering Fundamentals** | Microsoft **2025**

**LLM-Driven AI Engineering Bootcamp** | REVA University **2025**

**Prompt Engineering** | Infosys Springboard **2025**

**Machine Learning** | Rinex, NSDC / Skill India | Grade: A+ **2024**

## LEADERSHIP

**Core Member and Head of Media | Yantra IoT Club, REVA University** **2024 – 2025**

- Led promotional campaigns for 2 robotics events, ROBONEMESIS (microcontroller training) and a Follow Bot Competition, which drew 17K+ combined views on social media.
- Coordinated technical workshops and career panels with Amazon and Google professionals for 100+ students.